

Mercedes vs Alpine Mean Pit Stop Time

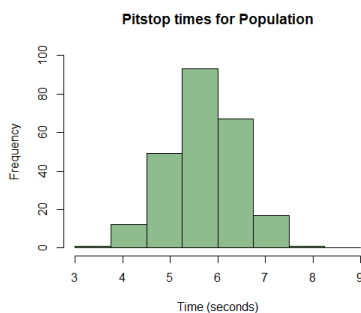
Introduction

Ten Formula 1 teams have recorded their fastest pitstop times (in seconds) at each Grand Prix throughout the championship. They are interested in the population mean of these pit stop times and, to see if there is a difference between the two population pitstop time means for the Mercedes and Alpine teams. The teams want to find out:

- What is the population mean pitstop time?
- Is there a difference between the two population pitstop time means for the Mercedes and Alpine teams?

Exploratory Analysis

The population mean pitstop time for the data collected was approximately 5.703 seconds.

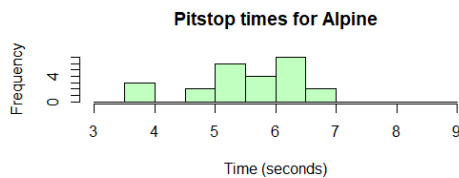
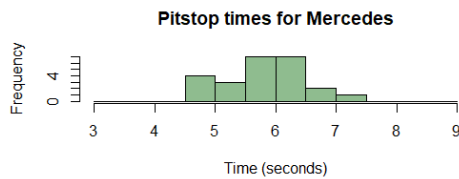


The distribution for pitstop times follows a normal distribution.

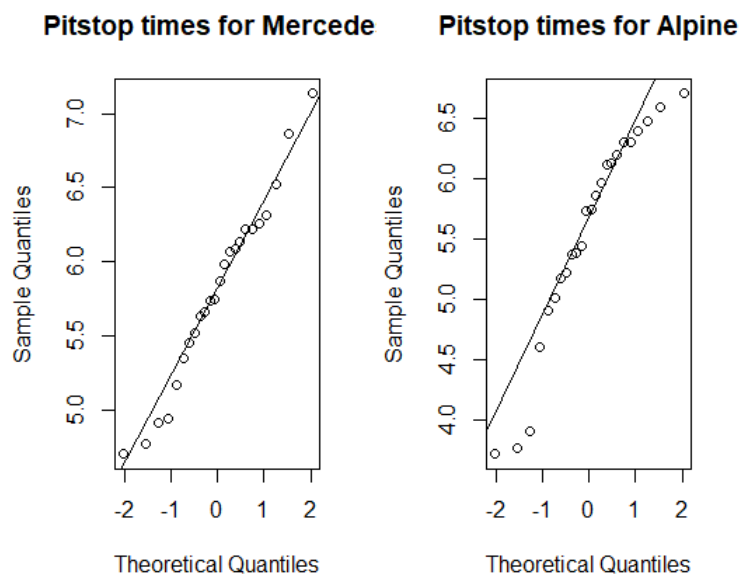
Sample mean of pitstop times for Mercedes and Alpine.

Team	Pitstop means
Mercedes	5.802042
Alpine	5.542125

This table is showing a small difference between the means of pitstop times for Mercedes and Alpine.



- the sample mean of pitstop times from Mercedes and from Alpine are approximately equal.
- the pitstop times from Mercedes and Alpine both approximately follow a normal distribution.



- the sample standard deviations of pitstop times from Mercedes and from Alpine are approximately equal.

Statistical analysis

For the population mean pitstop time, I calculated a 99% confidence interval, assuming that the data follows a normal distribution and the standard deviation was equal to 0.8 seconds. The 99% confidence interval for the population mean pitstop time is equal to [5.570364, 5.836394]. This interval is centred around the population mean.

Again, for the population mean pitstop time, I calculated a 99% confidence interval, assuming that the data follows a normal distribution and the standard deviation was unknown. The 99% confidence interval for the population mean pitstop time is equal to [5.575848, 5.830910]. This interval is centred around the population mean.

To answer whether there is a difference between the mean pitstop times from Mercedes and Alpine, a 99% confidence interval for the difference in means from the two populations is used. This is calculated assuming that both populations Mercedes and Alpine pitstop times follow a normal distribution which have unknown but equal variances.

The 99% confidence interval for the difference between the mean pitstop times for Mercedes and Alpine is [-0.3328659, 0.8526992]. This interval contains the value zero, so there is no statistically significant evidence at the 1% significance level to suggest the mean pitstops are different.

Conclusion

In the exploratory analysis for the population mean, we saw that the population mean was 5.703 seconds and followed a normal distribution.

Also in the exploratory analysis for the difference between Mercedes and Alpine pitstop times, we observed that there was no significant difference between the pitstop time means for both teams. This was backed up by the statistical analysis as the confidence interval contained the value 0, overall suggesting no difference in the mean pitstop times between Mercedes and Alpine.